

U-Jagd Manual

Player manual, version 1.2.0, as of 2026-09-25

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Quickstart

You command the ASW frigate F-217 and man nine stations. Your job: detect, track, classify and sink hostile submarines without losing the ship or harming neutral shipping.

Everything in this manual describes what the simulation actually models. Features that are not modelled are listed at the end of every station chapter.

Mission and win conditions

- **Win:** sink every assigned target submarine, or survive until the time limit (depends on the mission).
- **Lose:** own ship sinks, a civilian vessel is hit, the target escapes more than 150 NM from its start point, or time runs out on a sink mission.
- Time warnings arrive at 5, 2 and 1 minutes remaining: on a sink mission as a deadline, on a survive (convoy) mission as the time until the convoy is safe.
- When the mission ends, R restarts it with the same seed (an editor mission restarts itself) and M returns to the main menu. During a mission, Esc offers "Main menu (without saving)" next to save-and-exit.
- Score: 1000 per submarine sunk, 200 per unused torpedo, 500 for no civilian losses, up to 500 time bonus.

Stations

The ship is split into nine stations. Keys 1-9 select a station; pressing the number of the current station again cycles its pages.

1 Bridge	2 Sonar	3 Weapons
4 Damage	5 Operations	6 Radio
7 Engine	8 Helicopter	9 Electronic warfare

Each station shows only what its sensors and operators know. Sonar contacts are noisy bearings until a ping, TMA, buoy or cross-fix supplies range. No station shows "the truth".

F2 hands the current station to the autocrew; F3 shows which stations run automatically. Use it when you want to concentrate on one or two stations.

Controls

The game runs at 1280x720 and is designed for the uConsole keyboard and trackball. The trackball acts as a joystick: horizontal steers on the Bridge, vertical steps the station's main selection elsewhere. A mouse works too: wheel zooms charts, drag pans, click pins a tooltip.

Global keys (all stations):

Key	Action
Tab / Shift+Tab	Next / previous station
1 / 2 / 3 / 4	Bridge / Sonar / Weapons / Damage
5 / 6 / 7 / 8	Operations / Radio / Engine / Helicopter
9	Electronic warfare / ESM
Active station number	Press again to advance that station's page
Arrow keys	Station-specific selection or adjustment
+ / -	Telegraph (available at every station)
F1 / ?	Help (this display)
F2	Toggle Autocrew for the current station
F3	Open Autocrew overview
0	Weather & sonar analysis panel
F8	Tactical Unit Analyzer (read-only catalog)
F4	Simulation log view (live; requires simlog option; M: map of all contacts)
F9	Open local Commander LAN administration
F10	Options: language, fullscreen, audio, large text, tooltips, frame rate
F11	Event log and full telemetry overlay (station stays live)
N	Nations & units; in sonar: notch filter
S / L	Save / load (slots 1-5)
Alt+Enter	Fullscreen (all stations)
Q / E or mouse wheel	Chart zoom only on bridge, weapons, and helicopter
Drag	Pan chart (bridge, weapons, and helicopter)
K	Camera follow only on visible charts (drag disables it)
P	Plot mode on the Bridge/Weapons/Helo map and OPZ chart: marks, rulers, bearing lines, circles, DR lines (shared by all stations, saved)
M R B C D · Enter · Bksp	In plot mode: choose tool, place point with Enter or click (arrows move the cursor, Shift faster), delete the nearest object (Shift: all)
Esc	Cancel input or open exit dialog
R / M	After the mission ends: restart with the same seed / main menu

In the Remote Crew browser (Commander, F9) stations are operated with buttons; the keyboard helps with navigation:

Key	Action
1-9	Open one of your own stations
[/]	Previous / next own station
?	Open guide and station help
Arrow keys	Move within focused tab bar, track list or chart
Home / End	First / last entry of the focused list
+ / -	Zoom focused chart; Home fits the view
Arrow keys (chart)	Pan focused chart
Mouse over map	Details of the track, own ship, asset, wreck or chart position under the cursor
0	Open or close the weather & sonar analysis
Plot tool + click	Draw on the shared plot: pick a tool above the map, click once (mark, bearing line) or twice (ruler, circle, DR line)

The bottom status ticker shows the newest event and key telemetry; F11 opens the full event log and telemetry over the station without stopping it or taking its keys. F1 (or ?) opens the help overlay at any time. It has four categories: global keys, the current station (keys and standard procedure), sensors and tactics, and this manual.

Underwater acoustics in five minutes

- **Passive sonar gives bearing only.** Every contact starts as a line of bearing. Range comes from active ping, TMA, sonobuoys or a cross-fix.
- **Detection is signal against noise.** The passive sonar equation $SE = SL - TL - NL + DI - DT$ decides: the target's source level (louder = farther), transmission loss (spreading, absorption, layer and path losses), noise (own self noise plus wind, rain and nearby shipping), and the array gain. A contact is detected at $SE \geq 0$ dB. Wind and rain matter most when you run slow and quiet; at high own speed your own noise dominates.
- **Own speed is own noise.** Self-noise rises from 4 kn to 25 kn. Above 15 kn the propellers cavitate and passive range drops to about a third.
- **The layer (thermocline) bends sound.** Passive propagation is ray traced through the real sound-speed profile: above the layer a surface duct carries sound far, below it lies a shadow zone a few miles wide. In deep water the shadow is strong; in water of a few hundred metres, bottom bounce and multipath fill it beyond about 10 NM, so hiding below the layer mainly works close in. A ping into the shadow zone reaches only 35 % of its range.
- **Seabed and surface matter.** Rock and gravel reflect sound well, silt and mud absorb it; a rough sea scatters high frequencies. **Convergence zones** appear only where the water is deep enough for rays to turn back up.
- **Baffles:** own-ship noise is a soft 70 degree lobe astern of the hull array (and along the cable of the towed array). It masks, it does not blank.

```

      surface
~~~~~
frigate HMS )))          direct path
----- layer / thermocline ----- -6.5 dB across
      TAS below layer )))    submarine
                              (shadow zone for a shallow sensor)
_____ seabed

```

Navigation conventions

- Distances in nautical miles (NM), speed in knots (kn), depth in metres, frequency in Hz.
- Courses and bearings are true degrees: 000 north, clockwise. Charts are north-up.
- The game always runs in real time: one real second is one simulated second. There is no time acceleration and no pause; menus, help, options, save/load and losing window focus do not stop the simulation either.
- The ship turns at up to 0.8 degrees per second; speed changes take minutes. Plan manoeuvres early.

Your first patrol

1. Main menu: pick scenario 1 (Patrol) with 1 and start with Enter.
2. Bridge (1): press 1 again for the mission page, read objective and time limit.
3. Engine (7): select SLOW or 6-8 kn. Sonar (2): stream the towed array with Y.
4. Sonar BROADBAND page: look for a bright vertical trace; select it with the arrow keys and press Enter to follow it.
5. Classify with C, enable TMA with T, then turn the ship 30-60 degrees on the Bridge and hold the new leg for a few minutes.
6. When TMA or a ping gives range: release the contact to Operations (G), make it the target (M).
7. Weapons (3): set torpedo depth to the pinged target depth, fire with T.
8. Watch the sonar for an incoming torpedo; if one appears, go FLANK, turn away and stream the Nixie (V at Weapons).

Main menu, saving and options

- Menu: 1-4 scenario (4 = random with custom difficulty), W world mode, R new seed, F fullscreen, Enter start.
- S / L: save / load (slots 1-5). Saves are exact and deterministic: a loaded game continues identically.
- F10: options - language, fullscreen, audio, large text, tooltips, frame rate (30 or 60 FPS; 30 saves CPU on the uConsole and is the default), event log / telemetry as status ticker (default, more room for the station) or docked band, operator assistance off (default: raw data and manual analysis) or training (automatic line labels, blade-rate and catalogue/emitter candidates).
- F9: Commander / Remote Crew - lets browser clients on the LAN take stations.

1 Bridge

Purpose

The Bridge conns the frigate: course, speed and position relative to coast, contacts and threats. Every sensor depends on how the ship is driven. Fast and straight is loud and blind; slow, steady legs with deliberate turns make sonar and TMA work.

Displays and instruments

Page 1 (navigation) shows the chart and four panels; page 2 (press 1 again) shows the mission briefing.

CHART (north up)	COURSE / RUDDER
own ship + wake	course 045 > 080
published tracks	rudder 15 R
bearing lines / fixes	SPEED / ACOUSTICS
helicopter, buoys	HALF 10.0 kn
	38% own noise
	TACTICAL PICTURE
	threats, sensors,
	assets, weather
footer: <- -> course Up/Down telegraph U/V direct	

- **Course / rudder:** current course, ordered course, rudder angle and turn radius.
- **Speed / acoustics:** telegraph order, speed, own noise in percent and a CAVITATION warning above 15 kn.
- **Tactical picture:** observed threats (a heard torpedo launch transient or HF seeker pulses, a contact sonar classified as torpedo, or an air track flagged as a possible missile), sensor state (radar, TAS), assets (helicopter, buoys) and weather/day-night.
- **Chart:** synthetic chart depth and coastline, own ship, tracks published by the other stations. Wheel or Q/E zoom, drag pans, K follows own ship.

Bridge lookout reports

The bridge lookout (eye height 18 m, 7x50 binoculars) reports sightings in the event feed as AUSG lines, for example Bridge lookout: frigate (Admiral-Gorshkov-Fregatte) bearing 040°, 3.8 NM. The Remote Crew bridge lists the same reports under "Lookout reports". A contact is reported in up to three steps as it closes, each step once:

- **Sighted:** only the kind of object is clear (vessel, aircraft, small object on the surface).
- **Class:** the silhouette shows the class, for example merchant ship, warship, aircraft carrier, fishing vessel, speedboat, surfaced submarine, airliner or military aircraft.
- **Type:** close in the lookout names the type: cargo ship, tanker, passenger ship, tug, frigate, destroyer, corvette or combat aircraft; warships and military aircraft also with their class name. Merchant ships and airliners are identified by name, AIS or transponder, not by eye, so the lookout never reports their name or airliner type.

Class and type need a finer resolved silhouette than the sighting (Johnson criteria): by clear day a tanker is classed at about 7 NM and a frigate identified at about 4 NM, a speedboat is classed only inside 3 NM, and at night the type is made out only within a few cables. Fog, rain and sea state shorten every step.

The lookout also calls "land in sight" with the bearing of the nearest coast, and a torpedo wake with a banner. The class is held while the lookout keeps the contact. It appears in the chart and OPZ tooltips as "Lookout: ..." and is an observation only: it never sets the OPZ classification or the affiliation.

Keys

Key	Action
<- / ->	Rudder: change target course
Up / Down	Telegraph up / down
U	Enter target course directly (000-359)
V	Enter target speed directly (0-25 kn)
+ / -	Telegraph: engine order (ASTERN-STOP-SLOW-HALF-FULL-FLANK)
Chart	Mouse wheel: zoom, mouse drag: pan
Q / E	Zoom chart out/in
K	Camera follow on/off

The trackball steers the rudder while the Bridge is selected. U and V open direct numeric entry; the simulation keeps running while you type. Enter confirms, Esc cancels.

Standard procedure

1. Check mission page (1 twice): objective, time limit, area.
2. Transit at FULL, then slow to SLOW or 4-6 kn before the search area.
3. Search: steady legs of 3-5 min, then turn 30-60 degrees so TMA gains range.
4. Keep clear of the coast and shallow water; watch the chart depth.
5. Torpedo inbound: FLANK, turn away beam-on, order Nixie at Weapons.

Combat situation:

1. Torpedo reported: FLANK immediately, turn to put the torpedo bearing astern or on the beam.
2. Order the Nixie at Weapons (V); keep turning so the torpedo sees the decoy first.
3. Once clear, reduce speed below 15 kn so sonar can reacquire; never above 20 kn with the towed array out.

Pro tips

- TMA needs a real change of own velocity. A 30-60 degree turn followed by a steady leg of several minutes gives the best range estimate. Turning on the spot does not help.
- The turn rate grows with speed (about 0.75 deg/s at 10 kn, 1.2 at 16 kn, 1.9 at 25 kn), so the turning circle stays near 0.4 NM. A stopped ship cannot turn. Speed changes take minutes: about 90 s to 90 % of FULL, and a stop from FULL uses reverse propeller pitch and takes about 90 s. Start evasive turns early.
- In a hard turn at speed the ship heels outward a few degrees; in heavy seas the fin stabilizers damp the roll, but only with steerage way.
- In shallow water the hull squats: at 25 kn the draft grows by up to 3 m when the water is less than about five draughts deep. Slow down in shoal water.
- Sprint-and-drift: sprint at FULL to a new position, then slow to 4-6 kn and listen.
- Heavy flooding on one side gives a list and a steady yaw pull; correct with rudder.

- The ship cannot run aground onto land; it is pushed back, but shallow water limits the helicopter dipping depth (10 m bottom clearance).
- Water depth follows the tide (semi-diurnal, about 12.4 h, up to a few metres in shallow water). A passage that is safe at high water can ground the hull at low water; the HQ weather bulletin reports the current tide at the ship.
- Wind pushes the surface water: about 3 % of the wind speed, 20 degrees to the right of downwind, on top of the steady ocean current.

Not modelled

- No time acceleration, no pause and no autopilot waypoints for the frigate.
- No automatic torpedo identification: the alarm rests only on heard intercepts or the sonar operator's classification; a torpedo running silent outside seeker range can arrive unannounced. The lookout calls out only a visible wake.
- The lookout never reads a ship's name or flag and does not report navigation lights or day shapes.

2 Sonar

Purpose

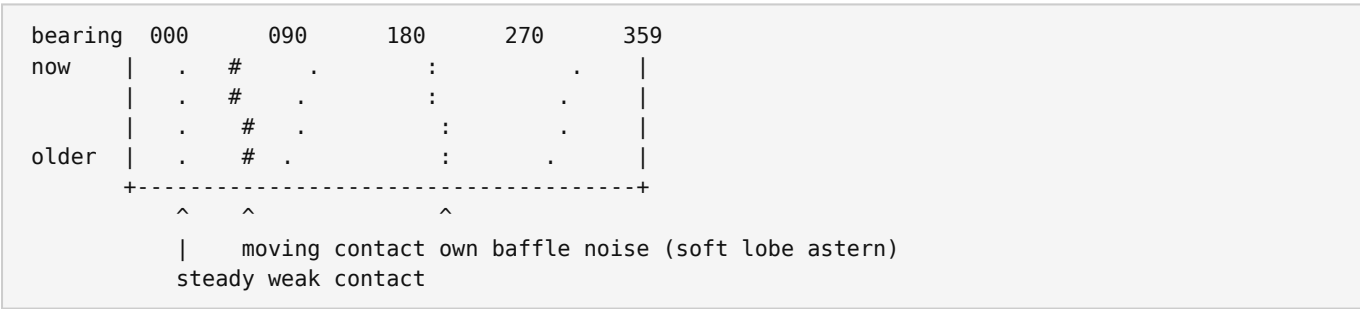
The sonar room is the main ASW sensor. It listens passively on the hull-mounted sonar (HMS) and the towed array (TAS), analyses signatures on LOFAR and DEMON, estimates target motion with TMA, measures the sound profile and, when ordered, transmits an active ping. It classifies contacts and releases them to Operations and Weapons.

Displays and instruments

The station has six pages. PgUp/PgDn (or 2 again) cycles them.

Page	Shows	Use it for
BROADBAND	Bearing-time waterfall	Detecting contacts and following their bearing
LOFAR	Frequency-time waterfall, 0-300 Hz	Tonal lines: machinery, shaft, blade signatures
DEMON	Modulation spectrum of the listening beam	Blade rate and shaft-rate hypotheses
TMA	Solution for the focused contact	Range, course and speed estimates
UMWELT / FUSION	Sound profile, layer, CZ, array comparison	Choosing TAS depth, spotting ghost contacts
ACTIVE	Stored echoes with age and error	Range and depth from pings

BROADBAND waterfall



Newest data is at the top. A straight vertical trace is a contact on a steady bearing; a slanting trace shows bearing rate. The history covers 20 s of fine data plus 4 minutes of long history (Shift+H selects 25/50/100 %). Brightness is relative received level, not range.

LOFAR

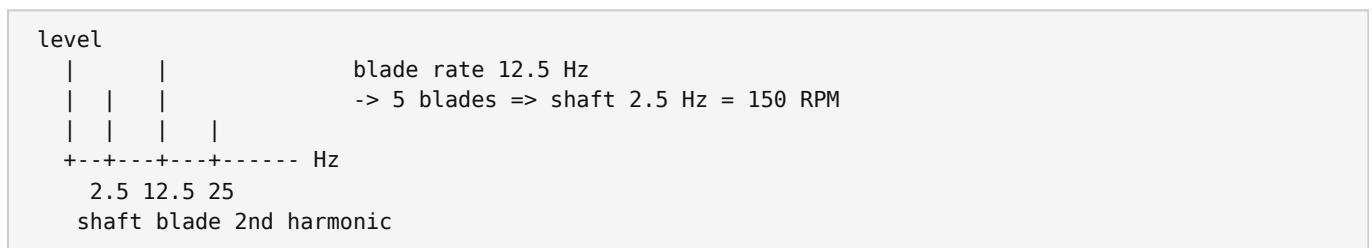
The x axis is frequency (0-300 Hz), time runs downwards. Bins are 1 Hz below 40 Hz, 2 Hz up to 100 Hz and 5 Hz above. A new line is added every 0.25 s; 80 lines are kept.

- Steady vertical lines are **tonals** (narrowband): generators, pumps, shaft lines. Several lines at integer multiples of one frequency are a harmonic family: put the white cursor on a line with Z/X (Shift: 10 Hz steps) and press K to mark it as fundamental; amber guides then show 2f, 3f and so on. K on the same frequency clears it.

- Own ship produces a shaft line at about $10 + 1.9 \times \text{own speed Hz}$. N notches it out.
- Space holds peaks so faint tonals stand out.
- The spectrum strip above the waterfall prints the frequency over every prominent line (interpolated between bins; with Space the held envelope). Where values would overlap, the stronger line keeps its label. The Remote Crew browser labels its spectra the same way.
- F selects the analysed band: FULL 0-300, LOW 4-80, SHAFT 8-55, MID 20-120 Hz. Ctrl+Z / Ctrl+X set the low / high band edge at the cursor for any band-, low- or high-pass; Shift+N puts an extra notch on the cursor frequency.
- Q selects the integration time: 2 s (the receiver's own FFT), 8, 16 or 64 s. Longer integration averages successive spectra so a weak steady tonal rises out of the noise, but a moving line smears. Shift+Q opens the vernier: 20 Hz around the cursor at the native 0.5 Hz resolution.
- The detail rail reads the level at the cursor. The strip labels lines automatically only with operator assistance set to training (F10).

DEMON

DEMON demodulates the broadband noise envelope of the listening beam. Propeller cavitation is modulated at the **blade rate** = shaft rate x number of blades.



The display shows measured modulation, not identity. After changing the bearing listen for at least a few seconds before judging. Count blades yourself: move the cursor (Z/X, 0.5 Hz) onto the shaft line and press K, then onto the blade line and press K again; the rail shows blades = blade rate / shaft rate (with the deviation from a whole number) and the shaft RPM. A third K clears both marks. Compare the result with the references in the contact analyser (F8). With operator assistance set to training (F10) the sonar also labels modulation lines, proposes RPM for 3-7 blades and ranks catalogue candidates.

TMA, environment and active

- **TMA** is yours. The page plots the bearings of the selected contact over time. Build a hypothesis: Z/X course (Shift: 1 degree), Ctrl+Z/Ctrl+X speed, Q/Shift+Q range on the newest bearing (Ctrl: 0.2 NM). The amber curve shows the bearings that hypothesis predicts, the dots at the foot the residuals (measured minus predicted). Good hypotheses leave residuals scattered around zero; a wrong course, speed or range leaves a trend. The rail shows the residual RMS, the systematic trend after averaging, the fit and the observability. Range is only observable after an own course change (at least 6 degrees, better 30-60): without one, K refuses. K accepts the hypothesis as the contact's TMA fix; it is dead-reckoned on its course and speed and ages out after 120 s, so refine and re-accept as bearings come in. Noisy bearings give a large range uncertainty even when the fit is good. TMA does not estimate depth. With operator assistance set to training (F10) an automatic solver proposal (it needs at least 4 bearings over 180 s) is drawn as a thin line and Shift+K copies it into the hypothesis. Sonobuoy bearings enter the track with the buoy as observer.
- **UMWELT / FUSION** shows the bathythermograph (E, 60 s cooldown): measured layer depth, sound-speed profile and convergence-zone bands, plus the HMS/TAS comparison. Bearings within 5 degrees confirm each other; 9 degrees or more apart are flagged as a possible ghost contact. The layer is not fixed: afternoon sun makes it shallower (about 8 m), strong wind mixes it deeper over hours, and internal waves move it a few metres. Repeat the BT after a few hours or a weather change. The

measured profile is the real temperature-driven sound speed (Mackenzie equation), so it drops below the layer.

- **ACTIVE** lists echoes of the last 120 s: bearing, range and depth (+/-12 m). A ping fix ages out after 120 s. W selects the pulse: **CW** (1 s tone) gives coarse range (about 0.1-0.3 NM) but its Doppler separates a moving target from seabed reverberation; **LFM** (100 Hz sweep) measures range to a few metres and gains 20 dB against noise, but a slow or stationary target stays inside the reverberation. The echo strength depends on the target's aspect (broadside about 15 dB stronger than bow-on) and size. Rocky ground reverberates far more than mud; charted wrecks return real echoes that no contact owns ("unassociated echo").
- **Hearing the echo:** every return is audible when it arrives, after its real two-way travel time (about 2.5 s per nautical mile of range), for the ship's sonar and the helicopter's dipping sonar alike. A CW echo is a soft tone on the carrier, an LFM echo a short sweep; the pulse is the one the ping was sent with. A strong echo stands out clearly, a faint one barely rises out of the reverberation hiss. In the Remote Crew browser the echo plays with the general sound (loud or faint).

Hull sonar versus towed array

	HMS (hull)	TAS (towed)
Passive range	1.0 x base	1.4 x base, minus 3 % per knot
Beam width	12 degrees	6 degrees
Bearing error	+/-6 degrees	+/-2 degrees
Self-noise	full	35 % of hull
Ping range	1.0 x	0.8 x
Handling	always ready	stream 360 s, recover 480 s, only at 3-12 kn, 30 s settle

- TAS depth 20-260 m (U/V in 10 m steps), limited to 260 m minus 4 m per knot of own speed. At 30 m or deeper and in the same layer as the target it gains another 25 %.
- Above 20 kn with any cable out the array suffers a permanent **FAULT**.
- The array heading lags the ship by about 45 s after a turn; its bearings are less reliable while it swings.

HMS noise lobe astern ^ \ ship / \ / ~~~~ ~~~~ 70 deg soft lobe	TAS noise lobe along cable ^ ship ===== cable === array (lobe points to ship)
---	--

Keys

Key	Action
Shift+A	Transmit active ping (cooldown; reveals position!)
Shift+B	Switch receiver array between HMS and TAS
Y	Deploy / retrieve TAS (only at 3-12 kn)
Page Up / Down	Broadband / LOFAR / DEMON / TMA / Environment / ACTIVE
2	Press 2 again to advance the sonar page
E	Bathymograph: measure local sound profile
W	Active pulse CW / LFM
U / V	Raise / lower TAS/VDS target depth by 10 m
R	Direct listening bearing: 000 to 359.9 degrees true
<- / ->	Bearing +/-0.5 degrees; Shift: 5, Ctrl: 0.1
Up / Down	Select contact for TMA and classification
Enter	Follow measured contact bearing / hold manually
J , / .	Receiver audio on/off lower/raise volume
A / B / H	Direct broadband / filtered / heterodyne audition
D	Toggle broadband/filtered audition
I / O	Lower / raise gain (3 dB)
Shift+I / Shift+O	Lower / raise sonar display contrast
Ctrl+I / Ctrl+O	Lower / raise sonar display black level
Shift+C	Cycle green / amber / cyan phosphor palette
Shift+H	Cycle displayed history depth: 25 / 50 / 100 percent
F	Select frequency band: wide / low / medium
N	Notch filter against own propulsion
K	Mark line at cursor (LOFAR fundamental, DEMON shaft/blade)
Z / X	LOFAR/DEMON frequency cursor (Shift: 10 Hz)
Ctrl+Z / Ctrl+X	Band-pass low / high edge at the cursor
Q	Integration time 2 (FFT)/8/16/64 s
Shift+Q	LOFAR vernier: 20 Hz at native 0.5 Hz
Shift+N	Notch at the cursor frequency
Shift+F	DEMON carrier band 200-800 / 400-1400 / 1000-2000 Hz
Ctrl+F	Heterodyne shift 400/700/1000/1200 Hz
X / Shift+X (BB)	Broadband/fusion: TAS side of the selected contact flip / Shift: confirm
Z / X (TMA)	TMA page: hypothesis course -/+ 5 deg (Shift 1 deg)
Ctrl+Z / Ctrl+X (TMA)	TMA page: hypothesis speed -/+ 1 kn
Q / Shift+Q (TMA)	TMA page: hypothesis range -/+ 1 NM (Ctrl 0.2 NM)

Key	Action
K / Shift+K (TMA)	TMA page: accept hypothesis as fix / Shift: copy solver proposal (training)
SPACE	LOFAR peak hold on/off
T	TMA for selected contact on/off
C	Classify contact (submarine / warship / biological / vessel / aircraft / torpedo)
G	Release / withdraw selected contact to CIC independently of classification
M	Set selected contact as target

Standard procedure

1. Go slow (4-8 kn) and select HMS; stream the TAS at 3-12 kn (Y).
2. Search BROADBAND for bearing traces; confirm tonals on LOFAR (N notch).
3. Select the contact (Up/Down), lock focus (Enter), check DEMON.
4. Classify (C); after a course change fit a TMA hypothesis (page TMA: Z/X, Ctrl+Z/X, Q) and accept it (K).
5. Need depth or exact range: ping (Shift+A), then release (G) and mark target (M).

Combat situation:

1. A launch transient, high-frequency sonar pulses, or a new loud broadband contact without tonals and with fast bearing drift may be a torpedo. Classify it as Torpedo (C) and report the bearing to the Bridge immediately.
2. Keep focus on the hostile submarine so the torpedo wire datum stays fresh.
3. Ping only when you need depth for the shot or the contact is about to be lost: the submarine hears a ping out to 60 NM and starts evading.

Pro tips

- Gain (I/O) changes display and audio only, not detection. Black level (Ctrl+I/Ctrl+O) and contrast (Shift+I/Shift+O) help faint traces stand out; Shift+C changes the phosphor colour.
- D or A/B/H choose broadband, filtered or heterodyne audition. Heterodyne shifts the low band up to about 700 Hz so low tonals become audible.
- The listening audio runs about one second behind the display so a busy moment never interrupts it. After steering the listening bearing the old beam fades into the new one after about a second; the stream is not cut.
- Put the TAS below the measured layer to hear deep targets; keep the HMS for shallow ones. Both arrays run in parallel.
- The TMA page shows the closing rate derived from the accepted solution: positive means the target is closing.
- If the TAS and HMS disagree by 9 degrees or more, treat the contact as a possible ghost (the display flags it) and turn to resolve it.
- The towed array is a line: it cannot tell a bearing from its mirror about the cable. A contact heard only on the TAS is marked "TAS left/right ambiguous" with its mirror bearing and does not feed TMA; the display shows the side you choose (starboard by default). Turn 20 degrees and watch both traces: the true one stays continuous, the ghost jumps (the status then reads "turn made - compare traces"). On the broadband or fusion page X shows the other side and Shift+X confirms the shown side; nothing is resolved for you. A wrongly confirmed side stays mirrored (the TMA residuals will show it; X reopens the choice). A hull-sonar bearing resolves the side by measurement. Bearings toward the cable ends (endfire) are also less accurate than broadside.

- Shift+F selects the DEMON carrier band (200-800, 400-1400 or 1000-2000 Hz): search the band where the cavitation noise is strongest. Ctrl+F sets the heterodyne shift (400/700/1000/1200 Hz) for listening to low tonals.
- In the contact analyser (F8) with a contact selected, Enter assigns the browsed catalog profile to that contact and Shift+Enter clears it. The assignment is your annotation: it is shown in the contact list and saved, and it never changes the contact's classification or weapon interlocks.
- A contact is lost 120 s after its last detection. Keep tracking weak contacts, or reacquire with a ping.
- Wrecks return real echoes without Doppler. A submarine lying still beside a charted wreck hides in that echo from a CW ping (750 m range cell); an LFM ping resolves about 8 m and can separate the boat from the wreck. Suspect every wreck the enemy could have reached.
- The bathythermograph (E) measures to the seabed, at most 1500 m. Only after a measurement does the weather & sonar analysis (0) show the layer, the shadow zone below it and a SOFAR channel.
- The sonar never names a torpedo or a submarine. It reports what it hears: a mechanical launch transient (heard out to 35 NM) or high-frequency seeker pulses (about 6 NM) as a bearing, held on the Bridge alarm for 60 s, and breaking-up noises when a hull sinks. The OPZ symbol of a sonar contact follows your classification only; an unclassified contact stays unknown.

Not modelled

- No dot stack or Ekelund range; TMA is the hypothesis/residual method above. T switches the solver behind the training aid.
- No selectable split-window normalisation (TPSW); use gain, black level and contrast instead.
- No hard blind baffle sector; own noise is a soft lobe.
- No variable-depth sonar separate from the TAS.

3 Weapons

Purpose

Weapons control turns a sonar contact into a firing solution. It launches the frigate's wire-guided torpedoes, manages the helicopter's stores (buoys, lightweight torpedoes), streams the Nixie towed decoy and releases the AA gun.

Displays and instruments

Page 1 (target) shows the chart with the selected contact, the torpedo depth and the fire-control readiness line. Page 2 (stores) lists tubes, reload timers, torpedo stock, Nixie state and helicopter stores.

The readiness line is checked top to bottom; the first failed check is shown:

```
BLOCKED: NO TARGET          select or take a contact (M)
BLOCKED: AFFILIATION ...    operator marked it FRIEND/NEUTRAL
BLOCKED: NO RANGE           ROE STD needs ping/TMA/buoy range
BLOCKED: NOT CLASSIFIED     classify as submarine or warship (Sonar C)
BLOCKED: NO TORPEDOES / NO TUBE READY / SALVO LIMIT
BLOCKED: WEAPONS ROOM DAMAGED
WEAPONS FREE                -> T or Ctrl+Enter
```

Torpedo run, seen from above:

```
frigate ==wire==> . . . . . /\ /\ /\ ( datum )
                   mid-course   snake search  seeker on
                   to datum      +/-15 deg     inside 1.2 NM
                   (wire update  -> homes on nearest
                   every 0.5 s)   candidate
```

- Frigate torpedo: 45 kn, 12 NM, two tubes, 60 s reload. Stock per mission is set by the scenario (default 6).
- Preset depth 10-300 m (default 60 m). A wrong depth is a miss: take depth from a ping, not from TMA.
- The wire updates the datum from the contact's observed position. Without updates it becomes STALE after 3 s and BROKEN after 12 s; the torpedo then continues to the last datum.
- The seeker homes on the nearest candidate: that can be a decoy, a whale or a merchant ship. A civilian hit ends the mission.
- Salvo doctrine SHOOT-LOOK-SHOOT: at most 2 own torpedoes running.

Keys

Key	Action
M	Set target (from sonar contacts)
Hold Up / Down	Torpedo depth (10-300 m)
<- / ->	Select sonar contact for targeting
T / Ctrl+Enter	Launch torpedo (ROE check)
H	Launch / recall HSP-5
B	Deploy sonobuoys (HSP-5 airborne)
D	Lightweight torpedo from HSP-5
V	Stream one finite towed acoustic decoy
Q / E	Zoom chart out/in
K	Camera follow on/off
F	Toggle AA gun fire release (withheld = never fires on raiders)

Standard procedure

1. Take target from sonar (M) or select a contact with Left/Right.
2. Check the readiness line: range fresh, classification, tubes, salvo limit.
3. Set torpedo depth to the pinged target depth (hold Up/Down).
4. Fire (T or Ctrl+Enter); keep the contact fresh so the wire datum updates.
5. Defence: stream the Nixie early (V); launch HSP-5 (H) for buoys and air torpedoes.

Combat situation:

1. Enemy torpedo reported: stream the Nixie at once (V). It lasts 600 s on a 0.2 NM cable; one ready, a second after 60 s. It runs at 10 m at 15 kn, deeper and closer astern when you slow down, and its cable parts above 25 kn. In a turn the cable lags behind.
2. Keep the counter-attack going: a fresh contact keeps the wire datum on the submarine.
3. With the helicopter airborne, a lightweight torpedo (D) can reach a distant contact faster than the ship's torpedo.

Rules of engagement

ROE	Requirement
STD (start)	Current range (ping, TMA or buoy) and classification submarine or warship
FREE	Classification only; without range the torpedo is aimed 10 NM down the bearing

HQ switches to FREE by radio after the first hostile submarine is sunk; the player cannot change ROE. A contact marked FRIEND or NEUTRAL in Operations, directly or through a fusion, can never be engaged.

Pro tips

- Fire from inside about 6-8 NM: at 45 kn the torpedo needs 8 minutes for 6 NM, and the submarine hears the launch out to 35 NM and starts evading.
- Aim the datum ahead of a moving target by keeping TMA running; the wire follows the observation, not the truth.

- Keep own speed below cavitation while guiding; losing the contact means losing the wire datum.
- The wire is a physical cable: it snaps if the ship runs faster than 20 kn or turns faster than 1.5 deg/s for about 5 s, when the ship-side spool (5 NM of own track) runs out, or when the torpedo has run 1.25 x its range.
- Torpedo range comes from its battery: at full speed it runs the catalogue range, hard manoeuvring throttles it and saves energy; when the battery is empty it coasts for a few seconds and is lost. Turning is slower right after launch (constant turning circle), and depth changes need a moment to build up.
- The warhead has a proximity fuze: it fires at the closest approach inside its radius, and the damage falls with distance (shock factor). A near miss can leave a submarine damaged but able to escape.
- A running torpedo is heard by the target through the sonar equation: quiet boats in calm water hear it from a few miles, rain and their own speed mask it.
- Only two Nixies per mission: stream the first when a torpedo is likely, keep the second for the next attack.
- Homing seekers lock on the loudest candidate and only switch when another is clearly (6 dB) louder. They ignore echoes without Doppler, so a hovering target is hard to find; a torpedo that overruns a decoy without a hull hit remembers it and re-attacks. Hostile submarine decoys fade as their battery drains; hostile warships stream their own decoys when they hear your torpedo launch.

Not modelled

- No depth charges, ASW rockets or ship-launched ASROC (ASROC is used only by friendly AI warships).
- No selectable torpedo search pattern and no manual enable point: the snake search and the 1.2 NM seeker switch-on are fixed (friendly ASROC payloads use a helix search at their splash point).
- One torpedo type for the ship and one for the helicopter; no selectable salvo doctrine.

4 Damage control

Purpose

Damage control keeps the ship afloat and the stations working after a hit. Three repair teams fight flooding and fire in nine compartments. Every compartment houses a station; a damaged compartment degrades it, a destroyed one disables it for the rest of the mission.

Displays and instruments

Page 1 is the ship schematic; page 2 lists details per compartment (flooding, fire, trend, teams on scene, heel).

bow				stern			
+-----+	+-----+	+-----+	+-----+	+-----+	+-----+	+-----+	+-----+
BRIDGE	SONAR	WEAPONS	OPZ	RADIO	ENGINE	FLIGHTDECK	
+-----+	+-----+	+-----+	+-----+	+-----+	+-----+	+-----+	+-----+
	HULL PORT				HULL STARBOARD		
+-----+	+-----+	+-----+	+-----+	+-----+	+-----+	+-----+	+-----+
state: OK -> DAMAGED / FLOODING -> DESTROYED				teams: 1 2 3			

- **FLOODING:** a hole below the waterline lets water in. The inflow follows the water pressure: fast at first, slower as the water inside rises towards the outside waterline. Rooms high in the ship (bridge) do not flood through a hole. At 70 % the compartment is DESTROYED.
- **DAMAGED:** the hole is patched; a small residual leak remains until a team pumps the room dry.
- **Patch kits:** plugging a hole uses one of 8 patch kits. Without kits a team can only pump against the open hole.
- **Hit location:** the torpedo's impact point decides the compartment; a close burst tears a bigger hole than a distant one. Missiles hit above the waterline and mostly start fires.
- **Fire:** grows with the room's fuel load (engine, flight deck and magazine burn fiercest) and is smothered by rising water. A room that stays hot for about 30 s ignites its neighbours. A flooded switchboard (sonar, operations, radio, engine) shorts and starts an electrical fire. A fire above 90 % in the weapons room cooks off the magazine: the room is destroyed and the neighbouring rooms are holed.
- **Stability:** floodwater adds weight, and loose water surfaces reduce the metacentric height (GM). The ship sinks when the floodwater exceeds its reserve buoyancy, and capsizes when GM is lost or the heel passes 35 degrees.
- **Heel:** off-centre floodwater lists the ship to that side and pulls it off course.
- **Steering gear and stabilizers:** the steering gear sits aft under the flight deck. If that compartment is destroyed, the rudder jams at its last angle until the room is repaired. A destroyed hull compartment on either side knocks out the fin stabilizers, so the ship rolls more in a seaway. Floodwater adds weight: the ship sits deeper and accelerates more slowly.

Station effects: a station loses capability continuously with flooding and fire in its room (sonar and radar range shrink gradually); a destroyed room disables it. A damaged engine room caps speed at 15 kn, a destroyed one at 8 kn; a damaged or destroyed weapons room blocks torpedo launches; a destroyed flight deck prevents helicopter launch and recovery; a destroyed operations room also disables ESM.

Keys

Key	Action
<- / ->	Select compartment
Up / Down	Select team 1-3 (unassigned)
Enter	Assign selected team to selected compartment
Backspace	Withdraw selected team
1-9	Always switch station; never assign a team
Click	Select compartment or its label; Enter assigns the selected team

On the uConsole the joystick buttons 1-3 assign team 1-3 directly to the selected compartment.

Standard procedure

1. After a hit, read the compartment plan: fire, flooding, destroyed.
2. Fire first: select the room (Left/Right) and a free team (Up/Down).
3. Assign with Enter; watch progress and heel on the detail page.
4. Then flooding in machinery and sonar spaces, then the hull.
5. Withdraw teams (Backspace) once a room is stable and reassign.

Pro tips

- Teams start in the operations room and need about 20 s per compartment to walk to their job; a team is only effective once it has arrived. Keep a team near the engine and weapons rooms.
- A team first patches the hole, then pumps. Patch kits are limited: spend them on rooms below the waterline, not on rooms that already stopped flooding.
- Two teams on one room halve the time. Teams cannot be assigned to a destroyed compartment.
- Fire in a room next to the engine or the weapons room is the most dangerous: it spreads into mission-critical spaces.

Not modelled

- No individual crew members or casualties; no longitudinal trim from flooding.
- No counter-flooding order; correct heel with repairs and rudder.

5 Operations / CIC

Purpose

Operations (OPZ / CIC) builds the tactical picture above the water: surface and air radar, AIS, released sonar and ESM bearings, manual fusion of reports, NATO affiliation and air defence. It hands designated tracks to Sonar and Weapons.

Displays and instruments

Page 1 is a full-height free chart with all published tracks; page 2 is the target page for the selected track. The ship-centred radar picture uses its own range scale (10/20/40/80/120 NM, PgUp/PgDn), independent of the chart zoom (wheel, down to 5 NM radius; drag pans; K follows). Own units come from the datalink, not from sensors: the ship, the airborne helicopter ("HSP-5 DL") and every own weapon under way, i.e. torpedoes from ship, helicopter or ASROC (T<n>), ASROC in flight and ESSM, each with a friendly symbol and a heading tick.

```
NATO frame colours (operator annotation, not truth)
  yellow = UNKNOWN   blue = FRIEND   green = NEUTRAL   red = HOSTILE

report sources:  radar  AIS  sonar(released)  ESM  HFDF  lookout
track list: ID  source  bearing  range  course/speed  age  class
```

- **Surface radar:** 30 NM, limited by the radar horizon (20 m mast) and target height; submerged submarines are invisible.
- **Air radar:** 100 NM for aircraft and anti-ship missiles (ASM).
- The antenna turns once every 4 s: a contact is updated only when the beam sweeps past it, and each sweep detects it with a probability that falls with range (50 % at the nominal range for a broadside ship; bow-on targets are seen later, fluctuating echoes can miss a sweep). Sea clutter grows with sea state (about -5 % at sea state 4, -25 % at 6) and rain attenuates the echo (-10 % surface, -20 % air); from sea state 5 measurement errors increase. Inside 3 NM the CIWS search/track radar holds an inbound missile continuously while CIWS is released.
- **AIS:** civilian ships broadcast course and speed every 2-10 s (3 min at anchor) and their name about every 6 min. The VHF receiver hears them only within line of sight (about 20 NM). A radar track of a civilian shows name and course only after the matching AIS report has been received; radar alone gives position only. Optional live AIS/ADS-B traffic is indistinguishable from simulated traffic.
- **Fusion:** mark 2-8 raw reports (Space) and fuse them (L) into one operator track; Shift+L dissolves it. A fusion whose reports come from exactly one sonar contact can be designated to Weapons; its classification counts for fire control unless Sonar has classified the contact itself, and its affiliation applies to that contact. A fusion lasts only while all its reports are current.
- **Suppression:** Delete hides a report locally; H shows suppressed reports again.

Keys

Key	Action
Up / Down	Select CIC track
C	Classify OPZ radar/HOJ reports or manual fusions
F	Set NATO affiliation
Shift+F	Cycle OPZ contact-domain filter
J	Enter a shared operator-visible track ID
Space / L / Shift+L	Mark raw reports and create/dissolve a manual fusion (Shift+L)
Delete / H	Suppress/restore locally; H manages suppressed reports
M	Pass CIC track to sonar/weapons
Page Up / Down	Radar range 10/20/40/80/120 NM
<- / ->	Select ASM track
E / Ctrl+Enter	Launch ESSM (VLS cell)
G	Deploy chaff (8 NM cone, cooldown)
R	Surface-search radar on/off (EMCON)
Shift+R	Air-search radar on/off (EMCON)
I	Toggle CIWS fire release (withheld = never fires on inbound ASMs)
Backspace	Clear all marked reports
Enter	Confirm attack after classifying a real contact hostile
K	Camera follow on/off

Standard procedure

1. Set radar range (PgUp/PgDn); switch radars for EMCON (R / Shift+R).
2. Select tracks (Up/Down), classify (C) and set NATO affiliation (F).
3. Merge duplicate reports: mark with Space, fuse with L.
4. Designate the track for sonar/weapons (M).
5. Missile threat: chaff (G), CIWS release (I), then ESSM (E).

Air defence sequence (missile inbound):

```
40 NM  ASM detected (air radar / ESM seeker bearing)
30 NM  ESSM envelope          -> E / Ctrl+Enter (2 fire channels)
 8 NM  chaff cone             -> G (40 % break-lock, short blindness)
1.5 NM  CIWS                  -> must be released with I
```

1. Air radar on (Shift+R), select the ASM track (Left/Right).
2. Chaff and manoeuvre first, then ESSM. Only 6 ESSM are loaded.
3. Keep CIWS released while missiles are inbound; withheld CIWS never fires.

Pro tips

- Radar is a transmission that hostile ESM can intercept. Switch radars off (EMCON) when stealth matters more than the air picture.

- The radar does not know what an air contact is. Threat evaluation flags an air track as a possible missile (ASM) only from its own measurements: faster than 300 kn at or below 150 m, or a jamming strobe; a low, fast attack aircraft can raise the same flag. The flag needs about a second of plots, CIWS and ESSM engage only flagged tracks, and HFDF fixes and unclassified sonar contacts carry no domain until you classify them.
- Anti-ship missiles skim at about 20 m (5 m in the last 5 NM): radar sees them only inside about 20 NM, and a jamming missile gives only a home-on-jam (HOJ) bearing until it burns through. Missiles fly inertially to their launch datum, then their seeker needs the ship inside its cone for 1.5 s before homing; attack aircraft pop up to about 300 m for a few seconds to lock their fire-control radar before each salvo (an ESM warning and an early radar contact).
- Chaff lays a cloud beside the ship that blooms in about 3 s and drifts with the wind; fire it early enough for the cloud to bloom. CIWS must first slew onto the missile and kills mostly in the last few hundred metres.
- Affiliation is your annotation. Marking a contact, or a fusion containing it, FRIEND or NEUTRAL blocks every torpedo shot on it.
- Chart symbols follow NATO style on the uConsole and on every Remote Crew map: the frame shows your affiliation (hostile diamond, neutral square, friend wide rectangle, unknown without frame), the inner glyph the observed domain.
- J gives a track a shared ID that the whole crew (and Remote Crew browsers) sees.
- Enter confirms an engagement against a live (real-world traffic) contact after you classified it hostile; nothing fires automatically on unclassified contacts.

Not modelled

- No sonobuoy management here: buoys belong to the helicopter station.
- No automatic track correlation across sensors; fusion is manual.
- No link-based air control of friendly aircraft.

6 Radio

Purpose

The radio room handles communications with HQ and HF direction finding (HFDF). HQ sends orders, weather bulletins and ROE changes by teletype. HFDF takes bearings on submarines that transmit on HF or run a snorkel mast, out to 120 NM, far beyond sonar range.

Displays and instruments

Page 1 lists current HFDF signals and the bearing log; page 2 is the teletype with HQ traffic.

HFDF SIGNALS	BEARING LOG
> HF-03 247.0 age 12 s	HF-03 247.0 from pos A t=12:04
HF-05 061.5 age 40 s	HF-03 239.5 from pos B t=12:08
	-> CROSS-FIX + error ellipse on chart
pos A *-----	
	-- X <- fix
pos B *-----/	

- Bearing error is +/-8 degrees for a ground wave and +/-16 degrees for a sky wave; signals older than 30 s can no longer be logged.
- Each signal shows its frequency and propagation. A submarine calling a distant shore station picks a high frequency by day (ground wave heard to about 95 NM) and a lower one at night (about 150 NM). Beyond the skip distance, several hundred NM away, the sky wave arrives instead.
- Logged lines and cross-fixes appear on the charts of Bridge, Weapons and Helicopter.
- A second bearing of the same signal gives a cross-fix if it is taken at least 1 NM away from the first and within 300 s.
- The teletype also carries the weather bulletin every 30 minutes and HQ messages (threat warnings, ROE FREE).
- At mission start HQ reports the threat. With **coarse** intelligence it gives only a rough bearing and range of one threat. With **exact** intelligence it also names every hostile unit type committed to the mission with its number (for example "1x Altmetall (Diesel, älter), 2x air raid wave with anti-ship missiles"), using the names in the unit analyser (F8); positions stay unconfirmed. Patrol always gets exact intelligence, Double hunt and Nuclear intercept coarse, and the free hunt lets you choose on its difficulty screen (last row, "HQ intelligence").

Keys

Key	Action
Up / Down	Select HFDF signal
Enter	Log bearing with own position

Standard procedure

1. Read the teletype on page 2: HQ orders, weather, ROE changes.
2. On HF activity select the signal (Up/Down).
3. Log the bearing (Enter); the line appears on the chart.
4. Move at least 1 NM and log a second bearing within 300 s.
5. Hand the cross-fix to Sonar/Bridge as a search datum.

Pro tips

- Take the two bearings from positions across the expected bearing line: the more they cross at right angles, the smaller the error ellipse.
- A submarine that transmits or snorkels is usually shallow and slow: a good moment to close in with the helicopter.
- Combine an HFDF bearing with a sonar bearing for a quick position estimate.

Not modelled

- No own radio transmissions or reports to HQ; no communication plan or crypto.
- No frequency tuning: HFDF monitors the whole HF band and lists the detected signals with their frequency.

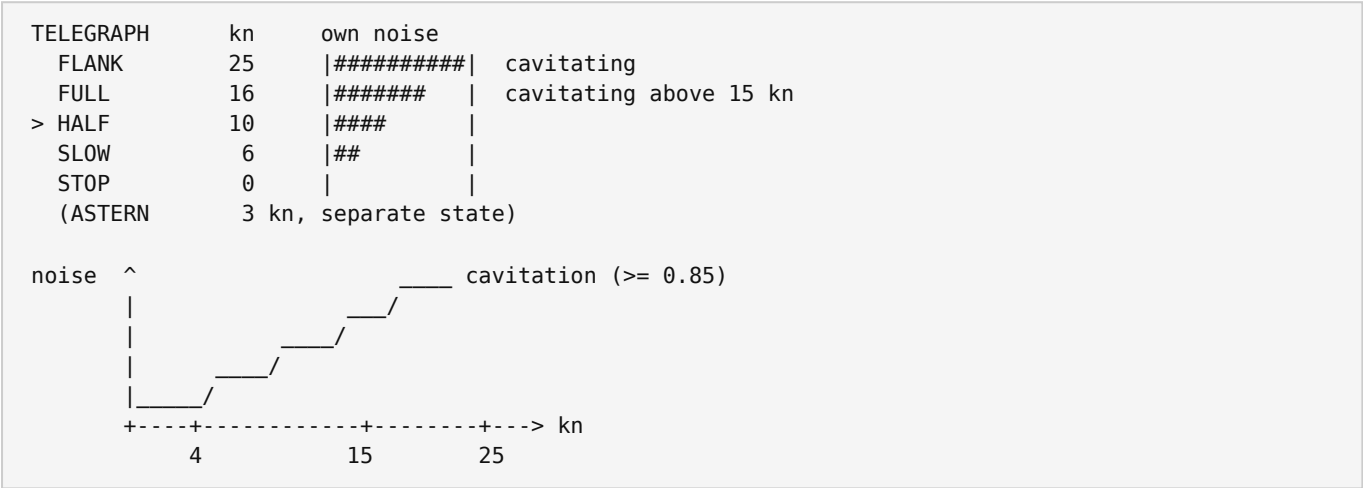
7 Engine room

Purpose

The engine room sets the propulsion order and manages the ship's acoustic signature. Speed is the most important trade-off in ASW: fast to reach a datum, slow and quiet to hear anything.

Displays and instruments

Page 1 is the engine telegraph with order, speed, shaft RPM and own noise; page 2 shows machinery systems, fuel and damage state.



- Own noise rises linearly from 4 kn to 25 kn. The propellers cavitate when the blade-tip speed is too high for the water pressure at the screws: in calm water from 15 kn, in heavy seas earlier when pitching lifts the stern. Cavitation raises noise to at least 0.85 and cuts passive sonar range to 35 %.
- QUIET mode reduces own noise to 65 % and limits speed to 12 kn.
- Shaft RPM follows the fixed-pitch propeller: about 5.8 rpm per knot at steady speed (146 rpm at 25 kn). While accelerating the control programme keeps the shaft at most about 11 rpm ahead of the present speed; when slowing down the pitch reverses and the shaft idles at 20 rpm. The own shaft line on LOFAR moves with speed.
- Machinery damage caps speed at 15 kn (damaged) or 8 kn (destroyed).
- Fuel burn follows the power the propellers deliver: at steady speed it grows with the cube of speed, accelerating and braking cost extra. A lighter ship (burnt fuel) accelerates slightly faster; floodwater makes it slower and deeper. Heavy seas add resistance and cost up to about 1 kn at FULL. With empty tanks the shaft stops and no engine order is accepted.

Keys

Key	Action
+ / -	Engine order (telegraph)
Up / Down	Telegraph up / down
A	Quiet mode QUIET/NORMAL
U	Enter target course directly (000-359)
V	Enter target speed directly (0-25 kn)

Standard procedure

1. Choose the order with Up/Down or +/-; V enters an exact speed.
2. Stay below 15 kn while listening: cavitation masks the sonar.
3. Switch QUIET (A) for the attack approach (max 12 kn).
4. Never exceed 20 kn with the TAS streamed.
5. Watch machinery damage: speed limits of 15 or 8 kn apply.

Pro tips

- Speed changes take minutes; order slow-down early before a listening leg.
- The towed array can only be streamed or recovered between 3 and 12 kn; at more than 20 kn with cable out it is lost.
- A hostile submarine hears you better than you hear it when you cavitate. Sprint only when you are far from the expected contact.

Not modelled

- No separate gas turbine / diesel plant selection and no individual shaft control.
- No refuelling at sea.

8 Helicopter deck

Purpose

The HSP-5 "Sea Lynx" extends the frigate's reach: it flies to a datum at 120 kn, drops sonobuoys, dips its own sonar and attacks with lightweight torpedoes, while the frigate stays quiet and out of torpedo range.

Displays and instruments

The station has four pages (8 again cycles them); it opens on page 3.

Page	Content
1 Status	Fuel, stores, weather limits, deck state
2 Mission	Chart with waypoint, buoys, contacts
3 Sonar	Dipping sonar and buoy contacts, depth, source
4 Acoustic	Listening: BROADBAND / LOFAR / DEMON of the dipping sonar or a passive buoy

frigate	waypoint (1-30 NM)
*----- 120 kn ----->	H hover + dip
buoy line (B): o ---- o ---- o	cable 15-300 m
PASSIVE: bearings;	))) dipping sonar
two crossing bearings (>= 10 deg) = fix	passive 18 NM
ACTIVE: range + bearing every 30 s	active 14 NM

- **Fuel:** 2 hours in forward flight; hovering (dipping) burns 1.3 times as fast. The helicopter returns automatically when only the 20-minute reserve remains. Running dry before landing loses the aircraft. In the hover the wind pushes it slightly downwind of its hover point.
- **Launch limits:** wind up to 32 kn, crosswind up to 22 kn, visibility at least 2 NM, sea state 5 or less, working flight deck, gusts up to 40 kn, cloud ceiling at least 300 ft, no severe icing, and a deck-motion window (roll within 8 degrees, pitch within 3.5 degrees). Light icing costs 20 % more fuel; the weather & sonar analysis (0) shows CLEAR, LIMITED or NO-GO. Landing also waits for such a window; turning into the sea reduces pitching.
- **Dipping sonar:** depth 15-300 m (default 75 m, at least 10 m above the seabed), passive 18 NM with +/-2 degrees, active ping 14 NM with 30 s cooldown. Dipping needs wind up to 30 kn, visibility of 1 NM and no icing.
- **Sonobuoys:** 5 per sortie, 8 NM range, 60 min battery; they drift with the current and a little with the wind. PASSIVE buoys give bearings (like DIFAR); ACTIVE buoys give range and bearing every 30 s (like DICASS).
- **Lightweight torpedo:** 2 per sortie, 55 kn, 12 NM, dropped from the helicopter's position towards the datum, no wire. The target must be classified as submarine.

Keys

Key	Action
H	Launch / recall HSP-5
Arrow keys	Adjust waypoint bearing and range
M	Set sonar contact as air-torpedo target
B	Deploy one sonobuoy at current position
Shift+B	Next sonobuoy mode PASSIVE / ACTIVE
T	Sensor source: dipping sonar / sonobuoys
F	Confirm / unconfirm selected helicopter contact
C	Classify contact (submarine / warship / biological / vessel / aircraft / torpedo)
G / Shift+G	Select dipping-sonar contact / release or withdraw it from CIC
Y	Lower / retrieve helicopter dipping sonar
U / V	Raise / lower dipping-sonar target depth
A	Transmit active ping from deployed dipping sonar
D / Ctrl+Enter	Drop lightweight torpedo
Q / E	Zoom chart out/in
K	Camera follow on/off
Acoustic: Page Up / Down	Acoustic page: Broadband / LOFAR / DEMON
Acoustic: <- / ->	Helicopter listening bearing -/+ 5 degrees
Acoustic: R	Reset listening bearing to automatic
Acoustic: T	Listening source: dipping sonar / passive buoys
Acoustic: J , / .	Receiver audio on/off lower/raise volume
Acoustic: I / O	Lower / raise gain (3 dB)
Acoustic: N	Notch filter against own propulsion
Acoustic: Shift+D	Cycle broadband / filtered / heterodyne audition
Acoustic: Shift+F	Cycle audition frequency band

Keys marked "Acoustic" apply only on the acoustic page (page 4).

Standard procedure

1. Check weather limits and fuel; launch HSP-5 (H).
2. Place the waypoint ahead of the suspected target (arrow keys).
3. Drop a passive buoy line (B); switch ACTIVE (Shift+B) for range.
4. Hover and dip (Y), set depth below the layer (U/V), listen or ping (A).
5. Classified submarine: mark target (M), drop torpedo (D); recall in time.

Attack sequence:

1. Localise with two passive buoys or an active buoy/dip ping until the contact has a fresh position.
2. Classify it as submarine (C) and set it as target (M).
3. Fly to the datum; drop the torpedo (D or Ctrl+Enter). Keep contact for a second drop if needed.

Pro tips

- Put the dipping sonar below the layer to hear deep submarines. The dip gauge shows the layer at the helicopter only once the lowered dome has passed through it; before that it shows only the charted water depth.
- Lay buoys ahead of the target's estimated track, not on top of the last datum.
- F confirms a helicopter contact; Shift+G releases it to Operations like a sonar contact.
- On the acoustic page, T switches the listening source between the dip and each passive buoy.
- Recall the helicopter in time (H): landing needs a working flight deck, and stores are not replenished between sorties.

Not modelled

- No buoy patterns (field, barrier) and no frequency channel management; buoys are dropped one at a time.
- No MAD (magnetic anomaly detector) and no radar on the helicopter.
- Only one helicopter.

9 Electronic warfare

Purpose

Electronic warfare (EloKa) listens passively for radar emitters (ESM) and, when ordered, jams them (ECM). ESM detects radars out to about 150 NM, far beyond own radar, without transmitting. It gives bearings and emitter parameters that point to a platform type - and warns when a missile seeker locks on.

Displays and instruments

Page 1 lists intercepts; page 2 shows the evidence for the selected intercept (frequency, PRF, modulation, candidates, correlation).

```
INTERCEPTS                                status  threat  band
> E-07  brg 312  9.3 GHz  PRF 2.4k  NEW     HIGH   X
  E-04  brg 045  3.0 GHz  PRF 1.0k  TRACK   LOW    S

EVIDENCE E-07
frequency 9.3 GHz  PRF 2400 Hz  modulation pulse-doppler
candidates (ranked, not identified):
  1. missile seeker  2. fire-control radar
correlation: compatible with CIC track T-12 (time/bearing)
```

- Bearing accuracy is about +/-3 degrees. Intercepts are bearings, not positions.
- A rotating search radar reaches the ESM antenna with its main beam once per revolution; its side lobes are heard only close in. The evidence page shows the peak signal level, a range estimate that assumes the power class of the best candidate (a wrong candidate gives a wrong range) and the measured antenna scan period.
- By default (operator assistance off, F10) the evidence page shows only the measured parameters and a library lookup: every emitter whose published frequency (and PRF) range contains the measurement, in name order, without score. Radar type, threat and range estimate are then yours to judge; C cycles the library in name order. With assistance set to training, candidates are ranked by frequency, PRF and modulation with a score, and radar type, threat and range estimate are filled in. A tie is not an identification.
- Correlation with radar or sonar tracks uses compatible time, bearing and observed position, never hidden identity.
- ESM runs from the operations compartment: a destroyed operations room disables it.

Keys

Key	Action
Up / Down	Select a visible passive ESM intercept
F / Shift+F / B	Cycle status / minimum-threat / frequency-band filters
C	Assign radar type and release current bearings to CIC; clearing the assignment withdraws release
J	Engage / release the directional ECM channel for the selected intercept
Shift+J	Cycle noise, RGPO, VGPO, or false-target ECM technique
A	Toggle automatic ECM prioritization and soft-kill coupling
M	Toggle local ELOKA contact sound

Standard procedure

1. Watch new intercepts; filter by status, threat and band (F/Shift+F/B).
2. Select an intercept and compare candidate radar types.
3. Annotate the radar type (C); this releases the bearing to CIC.
4. Missile seeker locked: jam (J), choose the technique (Shift+J).
5. Use auto ECM (A) under saturation; jamming is a transmission.

ECM techniques

Technique	Effect
Noise	Masks the victim radar with broadband noise
RGPO	Range-gate pull-off: drags the seeker's range gate away
VGPO	Velocity-gate pull-off: drags the Doppler gate away
False targets	Injects false returns

Automatic mode (A) picks targets and techniques and couples jamming with soft-kill (chaff) during a missile attack.

Pro tips

- Keep ESM running even under EMCON: it is completely passive.
- An intercept that changes from search to a high PRF pulse-doppler seeker at a steady bearing is a missile about to attack: warn Operations immediately.
- Assigning a radar type (C) releases the bearing to CIC; clearing the assignment withdraws it again.
- Jamming is a transmission. Use it deliberately, not continuously.
- Your own radar is heard too: a submarine at periscope depth (mast up, about 18 m) intercepts it on its ESM and can pass the bearing to other hostile units over its datalink. Deep boats neither hear radar nor receive the datalink.

Not modelled

- No communications intelligence (COMINT) or radio intercept; HF signals are handled by the radio room (HFDF).

- No decoy launchers other than chaff and no towed radar decoys.

Reference data

All values are the defaults of the current game version. Custom difficulty and missions can change stocks and time limits.

Units

Quantity	Unit
Distance, range	nautical miles (NM)
Speed	knots (kn)
Depth	metres (m)
Frequency	hertz (Hz)
Course, bearing	degrees true, 000 = north, clockwise
Time	simulation seconds; 1x only

Sensors

Sensor	Range	Accuracy / note
Passive sonar (base)	20 NM	bearing only; HMS +/-6 deg, TAS +/-2 deg
Active ping	18 NM (reference target, broadside-quarter aspect)	CW or LFM (w); range accuracy from pulse and SNR, depth +/-12 m; 30 s cooldown; heard to 60 NM
Dipping sonar	18 NM passive / 14 NM active	+/-2 deg
Sonobuoy	8 NM	60 min battery
Surface radar	30 NM (50 % per sweep)	4 s antenna revolution; radar horizon; no submerged contacts
Air radar	100 NM (50 % per sweep)	aircraft and missiles; jammers burn through close in
ESM	150 NM (main beam)	+/-3 deg bearing; level and range estimate
HFDF	120 NM ground wave at 15 MHz (about 95-150 NM by frequency)	+/-8 deg bearing (sky wave +/-16)
Lookout	12 NM surface, 5 NM surfaced sub, 20 NM air, 20 NM land	x0.25 (new moon) to x0.45 (full moon) at night; fog and sea state reduce; class at 2, type at 3.2 resolved cycles over relative size (tanker about 7/5 NM, frigate 5/4 NM, speedboat 3/2 NM by clear day)

Weapons and countermeasures

System	Data
Frigate torpedo	45 kn, 12 NM (battery), wire-guided (ship $\leq$ 20 kn, $\leq$ 1.5 deg/s, 5 NM spool), 2 tubes, 60 s reload, depth 10-300 m, proximity fuze
Helicopter torpedo	55 kn, 12 NM, 2 per sortie, no wire
Hostile torpedo	28 kn, 30 NM, homes from 3 NM
Nixie towed decoy	2 per mission, 600 s, 0.2 NM cable (10 m at 15 kn, deeper when slower, parts above 25 kn), 60 s reload
ESSM	6 missiles, 30 NM, 2 fire channels
CIWS	1.5 NM, 180 rounds, needs release; 115 deg/s slew, own track radar inside 3 NM
AA gun	240 rounds, needs release
Chaff	6 rounds, 8 NM, 40 % break-lock when bloomed; cloud drifts with the wind for 90 s

Own ship

Item	Value
Speed	4-25 kn; telegraph STOP 0, SLOW 6, HALF 10, FULL 16, FLANK 25 kn
Turn rate	about 0.075 deg/s per knot (1.2 deg/s at 16 kn); turning circle about 0.4 NM
Cavitation	from 15 kn in calm water, earlier in heavy seas; passive range $\times 0.35$
QUIET mode	noise $\times 0.65$, max 12 kn
TAS handling	3-12 kn, stream 360 s, recover 480 s, fault above 20 kn
TAS depth	20-260 m, minus 4 m per knot
Damage	9 compartments, 3 teams (about 20 s walk per compartment), 8 patch kits; sinks beyond reserve buoyancy, capsizes at 35 degrees heel or lost GM

Environment

Process	Model
Tide	M2 (12.42 h) + S2 (12 h), 0.4-1.4 m amplitude, larger in shallow water
Surface layer	seasonal base depth; about 8 m shallower in the afternoon; deepens in wind above 12 kn; internal waves +/-6 m
Sound speed	Mackenzie equation from the temperature profile (sea surface 8-18 deg C by season)
Current	steady field up to 1 kn plus 3 % of the wind, 20 deg right of downwind
Seabed	rock, gravel, sand, silt or mud; affects bottom reflection
Hazards	up to 64 charted wrecks and submerged rocks (tops at least 15 m deep), shown on every chart (wreck: hull line with masts, rock: asterisk; depth of the top when zoomed in, details in the tooltip); both raise the seabed within their footprint and are obstacles for the ship, submarines and weapons
Atmosphere	barometer 975-1025 hPa that falls ahead of rising seas; air temperature from the sea, season, day and cold northerly winds (below 0 deg C in winter storms: snow, icing); gusts; cloud ceiling; sun elevation with civil/nautical twilight; moon phase
Rain lens	rain freshens the top few metres (up to -1 PSU, mixed away by wind) and lowers the surface sound speed
SOFAR channel	an interior sound-speed minimum (about 400-500 m below the surface layer) exists only in deep enough water

Weather & sonar analysis (key 0)

Key 0 opens a full-screen analysis panel over any station (0 or Esc closes it; the simulation keeps running). In the web client every station opens it with 0 or from the workstation menu.

- **Environment:** time, daylight (day, civil or nautical twilight, night), moon phase, weather and precipitation, visibility, wind with gusts and Beaufort force, sea state, barometer with its 3-hour tendency (rising, steady, falling, falling rapidly), air and sea temperature, cloud ceiling and icing. A rapidly falling glass below about 1004 hPa gives a storm warning. The weather system changes by at most one sea state per hour, so the barometer moves faster than a real one.
- **Weather effects:** sun (strong layer), wind (deeper mixed layer) and rain or snow (fresher surface water, rain noise) light up while they act.
- **Helicopter flight weather:** CLEAR, LIMITED (within 80 % of a limit, or light icing) or NO-GO, with wind, gusts, crosswind, visibility, ceiling, sea state, deck roll and pitch, icing and whether dipping is possible.
- **Ocean profile:** appears only after the sonar has taken a bathythermograph (Sonar E): measured sound speed over depth, the layer, a SOFAR axis if present, nine sound rays from the hull sonar to 20 NM and the shadow zone below the layer (red) where the hull sonar hears little. The measurement is marked stale after 30 min or 10 NM.

Chart plot tools (key P)

The crew keeps one shared grease-pencil plot. Every station and every Remote Crew browser sees the same drawing, and it is saved with the game. It is the crew's own drawing: nothing in it comes from a sensor, and it never changes the simulation.

- **Opening it:** press P on the Bridge, Weapons or Helicopter map or on the OPZ chart. A cursor appears on own ship. Arrow keys move it (Shift: faster), or click on the chart. Enter sets a point, Esc cancels a

started object and then ends plot mode, and P also ends it. A hint bar at the top of the chart shows the active tool and keys on the left and the cursor's bearing and distance from own ship on the right.

- **Tools:** M mark (one point); R ruler (two points, shows bearing and distance); B bearing line from own ship through the cursor (own position and time are stored, so the line stays where it was laid); C circle (centre, then a point on the radius, at most 200 NM); D dead-reckoning line (start point, then a point in the direction of travel, then type the speed 0-60 kn). The DR line moves on with time and shows its CPA to own ship's present course and speed.
- **Erasing:** Backspace deletes the object nearest the cursor. Shift+Backspace clears the whole plot.
- **Labels:** objects are numbered M1, R2, B3 and so on. In the web client you can type a label before drawing or rename an object in the list under the map.
- **Web client:** choose a tool above the map, then click once (mark, bearing line) or twice (ruler, circle, DR line). "Plot track bearing" lays the selected track's measured bearing from its observer position.
- **Limits:** at most 64 objects and 24 characters per label.

Opposing submarines

Class	Quietness	Max depth	Torpedoes
Diesel (older)	0.75	200 m	4
AIP (modern)	0.85	250 m	5
Nuclear attack	0.92	400 m	8

Submarines evade for 240 s after hearing a ping or a torpedo, may launch a decoy, lie in wait, snorkel (detectable by HFDF and ESM) and sometimes ping from 15 NM or less. Near the frigate a boat may instead creep to a charted wreck within 8 NM and lie still on the bottom beside it for 15-30 minutes.

Submarine physics: the hull accelerates toward an ordered speed (no instant sprints); hydroplanes need speed (below about 4 kn depth changes are slow); radiated noise rises about 12 dB per doubling of speed and jumps when the screw cavitates, and the cavitation speed rises with depth; a torpedo launch makes an 8 s transient; a badly flooded boat blows ballast once and rises fast and loud; operating below test depth fatigues the hull, and 1.5 x test depth crushes it; a lurking boat holds its position against the current. Submarines sense like you do: passive bearings from their own sonar, a range only after their own TMA legs (a few minutes), ESM only with the mast up, the datalink only at mast depth or snorkelling, and a torpedo alarm takes the crew a few seconds (2-15 s) before the boat evades. Surface ships lose top speed in heavy seas (small ships more).

Mission and scoring

- Scenarios: 1 Patrol, 2 Double hunt, 3 Nuclear intercept, 4 Random (custom difficulty). User missions start from the Mission Editor (F5 in its browser).
- Win: all targets sunk, or survive the time limit. Lose: own ship sunk, civilian hit, target 150 NM from its start, or time out.
- Score: 1000 per sunk submarine, 200 per unused torpedo, 500 without civilian losses, up to 500 time bonus.

Glossary

Term	Meaning
ASW	Anti-submarine warfare
HMS / TAS	Hull-mounted sonar / towed array sonar
LOFAR	Low-frequency analysis and recording: frequency-time waterfall
DEMON	Demodulated noise: reveals blade and shaft rate
TMA	Target motion analysis from bearings
BT	Bathymograph: measures the sound-speed profile and layer
CZ	Convergence zone
Datum	Last estimated target position used for weapons and search
CIC / OPZ	Combat information centre / Operationszentrale
ESM / ECM	Electronic support (passive intercept) / countermeasures (jamming)
HFDF	High-frequency direction finding
EMCON	Emission control: radars off
ROE	Rules of engagement

Screen abbreviations

When a full label does not fit the 1280x720 screen, the station shows its catalogue abbreviation instead of cutting the text off. The F11 log and tooltips always show the full wording.

Short	Meaning
CRS/SPD	Course / speed
NOISE, CAV	Own radiated noise in %, cavitating
SS/LAYER	Sea state / measured layer depth (BT)
FLOOD	Mean flooding
TORP, VLS/CHAFF	Torpedoes left, VLS cells / chaff reload
HELO/ROE, HGR	Helicopter state / rules of engagement, hangar
BRG, G, N	Bearing, gain, notch
BB, FILT, HET	Broadband, filtered, heterodyne audition
STOW, DEPLOY, RECOV, OUT, STAB	Towed array stowed, deploying, recovering, streamed, stability
RDY, N/RDY	Ready, not ready
UNK, FRD, NEU, HOS	Affiliation: unknown, friend, neutral, hostile
SFC, SUB, AIR, MSL, TRP	Domain: surface, subsurface, air, missile, torpedo
RDR S/A	Radar surface / air
D, W, CD	Dipping sonar depth, water depth, ping cooldown
B-rate, Tgt	Bearing rate, target